# Calcurion Maths Worksheets

A-Level Further Mathematics

**Topic:** Further Mechanics 1 – Centres of Mass

Medium & Hard • 25 Questions

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## Section A – Medium (Q1–15)

**1.** Find the centre of mass of three particles of masses $1, 2, 3\,\text{kg}$ at positions 0, 2, 6 on a line. **[3]**

**2.** Find the CM of a uniform rod of length $2l$. **[1]**

**3.** Find the CM of a uniform triangular lamina. **[2]**

**4.** Find the CM of a uniform semicircular arc of radius $r$. **[3]**

**5.** Find the CM of a uniform semicircular disc of radius $r$. **[3]**

**6.** A uniform square lamina of side $2a$ has a square of side $a$ removed from one corner. Find the CM of the remainder. **[5]**

**7.** State the formula for CM of a system of particles: $\bar{x} = \frac{\sum m_i x_i}{\sum m_i}$. **[2]**

**8.** Find the CM of a uniform solid hemisphere of radius $r$. **[3]**

**9.** A composite body consists of a uniform rod of mass $2m$ and length $2l$ with a particle of mass $m$ at one end. Find the CM from the other end. **[3]**

**10.** Find the CM of an L-shaped lamina made of two rectangles $6 \times 2$ and $2 \times 4$. **[5]**

**11.** A triangular framework of three uniform rods each of length $2a$ and mass $m$. Find the CM. **[4]**

**12.** Explain why the CM of a uniform body can lie outside the material (give an example). **[2]**

**13.** Find the CM of a uniform wire bent into a quarter circle of radius $r$. **[3]**

**14.** A system of particles has total mass $M$ and CM at $\mathbf{r}$. A particle of mass $m$ is added at $\mathbf{a}$. Find the new CM. **[3]**

**15.** Find the CM of a uniform rectangular lamina $a \times b$. **[1]**

## Section B – Hard (Q16–25)

**16.** A uniform disc of radius $2a$ has a circular hole of radius $a$ cut with centre at distance $a$ from the disc centre. Find the CM of the remainder. **[6]**

**17.** A solid consists of a uniform cone of height $h$ and base radius $r$ joined at its base to a hemisphere of radius $r$. Find the CM from the vertex of the cone. **[6]**

**18.** Use integration to find the CM of a uniform triangular lamina with base $b$ and height $h$. **[5]**

**19.** A lamina is formed by removing a triangle from a rectangle. Set up the CM calculation. **[5]**

**20.** Find the CM of a uniform solid cone of height $h$. **[4]**

**21.** Three uniform rods form a right-angled triangle with sides $3, 4, 5$. Find the CM. **[5]**

**22.** A particle of mass $m$ is attached to a uniform rod of mass $M$ and length $2l$ at one end. The system pivots about the other end. Find the period of small oscillations (compound pendulum idea). **[5]**

**23.** Find the CM of the region under $y = x^2$ from 0 to 1 (plane lamina). **[5]**

**24.** A body is suspended so that it hangs in equilibrium. What does this tell you about the CM? **[2]**

**25.** A uniform square plate has CM at its centre. A circular hole is cut. Describe how the CM moves. **[3]**

*End of Worksheet* • *Show all working clearly.*